PCD_Assignment03 Report Analysis

Rameyza Charisa Putri Primantoko
25/569972/PA/24024

In this assignment, image segmentation and morphological processing were applied to an image containing several coins on a black background. The main purpose was to separate the coins from the background and compare the results before and after morphological processing. Otsu's thresholding was used for segmentation, while dilation, erosion, opening, and closing were used to process the binary image. All methods were implemented manually without using built-in image processing functions.

The first step was converting the original image into grayscale. This was done to simplify the image because grayscale only contains intensity values ranging from 0 to 255. After that, Otsu's thresholding was applied to determine the threshold value automatically. The image was then converted into binary form, where white pixels represented the coins and black pixels represented the background. Since the coins have different colors and patterns, some darker areas inside the coins may also appear black in the segmentation result. This shows that Otsu's method can separate objects based on brightness, but the results are not always perfect.

After segmentation, morphological operations were applied using a^ 3×3 structuring element. The first operation was dilation, which expands the white regions by adding pixels around the object boundaries. This makes the detected coins appear larger and can reduce small gaps. However, dilation may also connect objects that are close to each other. The second operation was erosion, which works in the opposite way by removing pixels from the boundaries. Erosion makes the foreground regions smaller and can remove small white details, although some important parts of the objects might also disappear.

The next operation was opening, which combines erosion followed by dilation. Opening is useful for removing small white noise while keeping larger objects. Compared to erosion alone, opening can restore some of the original object size. Closing was also applied by performing dilation followed by erosion. This operation helps fill small black holes and narrow gaps within the white regions. In the coin image, closing can help make the detected coin regions more complete, especially when some dark patterns appear as holes after segmentation.

The results were compared by observing the images and calculating the number of foreground pixels after each operation. Dilation generally increases the number of white pixels, while erosion decreases it. Opening and closing also change the foreground pixels, depending on the details and gaps present in the image. Since a 3×3 structuring element was used, some differences may be small when viewing the entire image, but they can still affect the object boundaries.

Overall, this assignment shows that morphological processing can be used to modify and refine binary segmentation results. Each operation has a different effect, so the most suitable method depends on the image and the expected result. Morphological processing can improve certain parts of the segmentation, but applying it without considering the object shapes may also remove useful details or connect objects that should remain separate.